

AVERAGE & 90th PERCENTILE WATER YEAR

Simulation of average and 90th percentile rainfall water years (WY) have historically been used in Los Angeles County (County) for water quality planning (LARWQCB 2014). To support future planning, an analysis was completed for the updated meteorological conditions in WMMS 2.0. The regional rainfall analyses below computed statistics for both the historical rainfall record (WY 1991 to 2020) and the most recent 10-years of rainfall (WY 2011 to 2020). The *WMMS 2.0 Phase I Report: Model Calibration and Validation* provides a full description of how these timeseries were developed (available on the [WMMS 2.0 Repository](#)).

First, a representative timeseries was developed for each of the major watersheds in the region using the 906 precipitation zones (developed as part of the *WMMS 2.0 Phase I* process; see Figure 1). This was completed by area-weighting timeseries for each precipitation zone across its respective watershed. Then, three metrics were used to determine average and 90th percentile WY:

- Metric 1- total annual rainfall,
- Metric 2- peak daily rainfall, and
- Metric 3- intensity (total annual rainfall divided by annual number of wet days). Wet days are those having 0.1 inches of precipitation or greater plus the following 3 days.

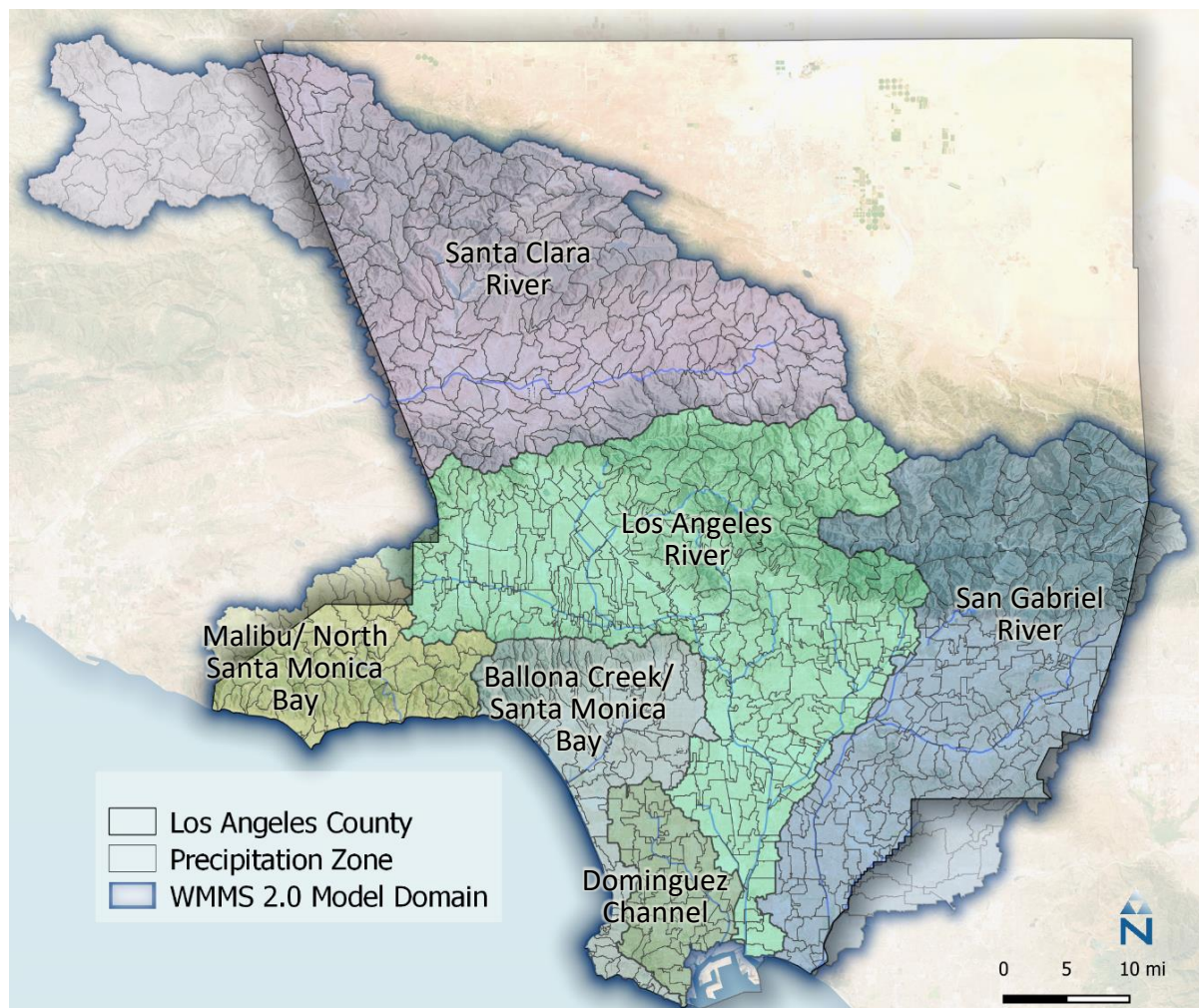


Figure 1. Major watersheds in the model domain and the 906 WMMS 2.0 precipitation zones.

Values for the three metrics for each WY for the most recent 10-years of data are presented in Table 1. Additionally, average and 90th percentile values from the historical rainfall record are listed at the bottom of the table and compared to each WY. To determine the average and 90th percentile for the most recent 10-year period, all metrics are equally weighted ($\frac{1}{3}$ each) within each WY and ranked against the long-term average and 90th percentile values. For some watersheds, metrics for two or more years were close to the historical metrics. For maximum coherence across the region, the WY that was most consistently close to the historical values across most watersheds was selected in those instances. Table 1 shows 2015 selected as the average WY (peach) and 2019 selected as the 90th percentile WY (blue).

Table 1. Values for rainfall analysis metrics 1, 2, and 3 for the last 10-years of rainfall data compared to the historical rainfall record (WY 1991 to 2020)

Water Year	Dominguez Channel			Los Angeles River			Ballona Creek/ Santa Monica Bay			San Gabriel River			Santa Clara River			Malibu/ North Santa Monica Bay		
	Metric 1	Metric 2	Metric 3	Metric 1	Metric 2	Metric 3	Metric 1	Metric 2	Metric 3	Metric 1	Metric 2	Metric 3	Metric 1	Metric 2	Metric 3	Metric 1	Metric 2	Metric 3
2011	2.13	17.58	0.58	3.99	25.16	0.70	3.29	21.81	0.66	3.99	29.32	0.76	4.66	23.99	0.73	4.56	24.79	0.79
2012	1.18	8.40	0.48	1.68	10.99	0.66	1.47	9.25	0.59	1.79	12.61	0.62	1.81	10.38	0.64	1.86	11.53	0.78
2013	0.93	5.79	0.37	1.04	8.28	0.40	1.10	7.20	0.44	1.12	9.69	0.42	0.93	6.43	0.36	1.27	8.10	0.43
2014	1.18	4.54	0.47	3.18	7.09	0.69	2.28	5.58	0.63	2.97	9.18	0.60	2.67	7.28	0.68	2.37	5.92	0.72
2015	1.44	10.02	0.51	2.08	12.08	0.63	2.02	10.57	0.59	2.10	14.45	0.63	2.05	11.12	0.61	1.95	11.25	0.66
2016	0.99	6.01	0.45	1.82	10.07	0.57	1.46	7.22	0.54	1.85	11.42	0.57	1.39	8.62	0.54	1.61	8.12	0.55
2017	3.17	18.36	0.65	3.00	21.87	0.65	2.50	19.04	0.63	3.27	25.43	0.72	2.87	17.91	0.57	3.38	21.14	0.67
2018	1.14	4.29	0.37	1.85	8.09	0.64	1.50	6.26	0.58	2.17	8.48	0.57	1.62	8.10	0.64	2.06	8.16	0.72
2019	2.25	17.70	0.68	3.02	25.22	0.77	2.30	21.54	0.76	3.93	29.21	0.85	2.59	21.40	0.68	2.41	22.73	0.80
2020	1.83	13.17	0.66	2.28	17.56	0.70	1.64	14.43	0.65	2.95	20.30	0.73	1.84	16.82	0.65	1.68	15.60	0.68
Average ⁴	1.62	10.59	0.52	2.40	14.64	0.64	1.96	12.29	0.61	2.61	17.01	0.65	2.24	13.20	0.61	2.31	13.73	0.68
90 th Percentile ⁴	2.34	17.77	0.66	3.26	25.16	0.71	2.57	21.57	0.67	3.94	29.22	0.77	3.05	21.66	0.69	3.50	22.93	0.79

Note: shading indicates years in record that most closely match the long-term rainfall statistics.

¹ **Metric 1:** Peak daily rainfall.

² **Metric 2:** Total annual rainfall.

³ **Metric 3:** Intensity (total annual rainfall divided by number of wet days)

⁴ Indicates the average and 90th percentile values from the 28-year precipitation record.

Table 2 summarizes the selected WYs for the average and 90th percentile values considering the analysis above. Also shown in Table 2 are the years that would be selected if the average and 90th percentile values could be compared to the entire historical record (WY 1991 to 2020). Analysis with WMMS 2.0 could either use the recent 10-year analysis of the long-term analysis, depending on the application.

Table 2. Recommended WY selection for average and 90th percentile values based on recent 10-year period and full period of record

Watershed	Recent 10-years		Historical 28-years	
	Average Water Year	90 th Percentile Water Year	Average Water Year	90 th Percentile Water Year
Dominguez Channel	2015	2019	2015	1995
Los Angeles River	2015	2019	2006	1995
Ballona Creek/Santa Monica Bay	2015	2019	2015	1995
San Gabriel River	2015	2019	2010	1995
Santa Clara River	2015	2019	2015	1995
Malibu/North Santa Monica Bay	2015	2019	1990	1995

To simulate the selected WYs, users can download the full 10-year calibration boundary conditions files (.pre and .air) on the WMMS 2.0 Repository and adjust model output start and end date in LSPC to match the intended year.

REFERENCES

LARWQCB (Los Angeles Regional Water Quality Control Board). 2014. Basin Plan for the Coastal Watersheds of Los Angeles and Ventura Counties. Sept. 11, 2014. Retrieved January 2020 from https://www.waterboards.ca.gov/losangeles/water_issues/programs/basin_plan/basin_plan_documentation.html